Signal Operations: Shifting, Folding & Scaling

Notes — 20th July 2026

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1. Shifting Operation

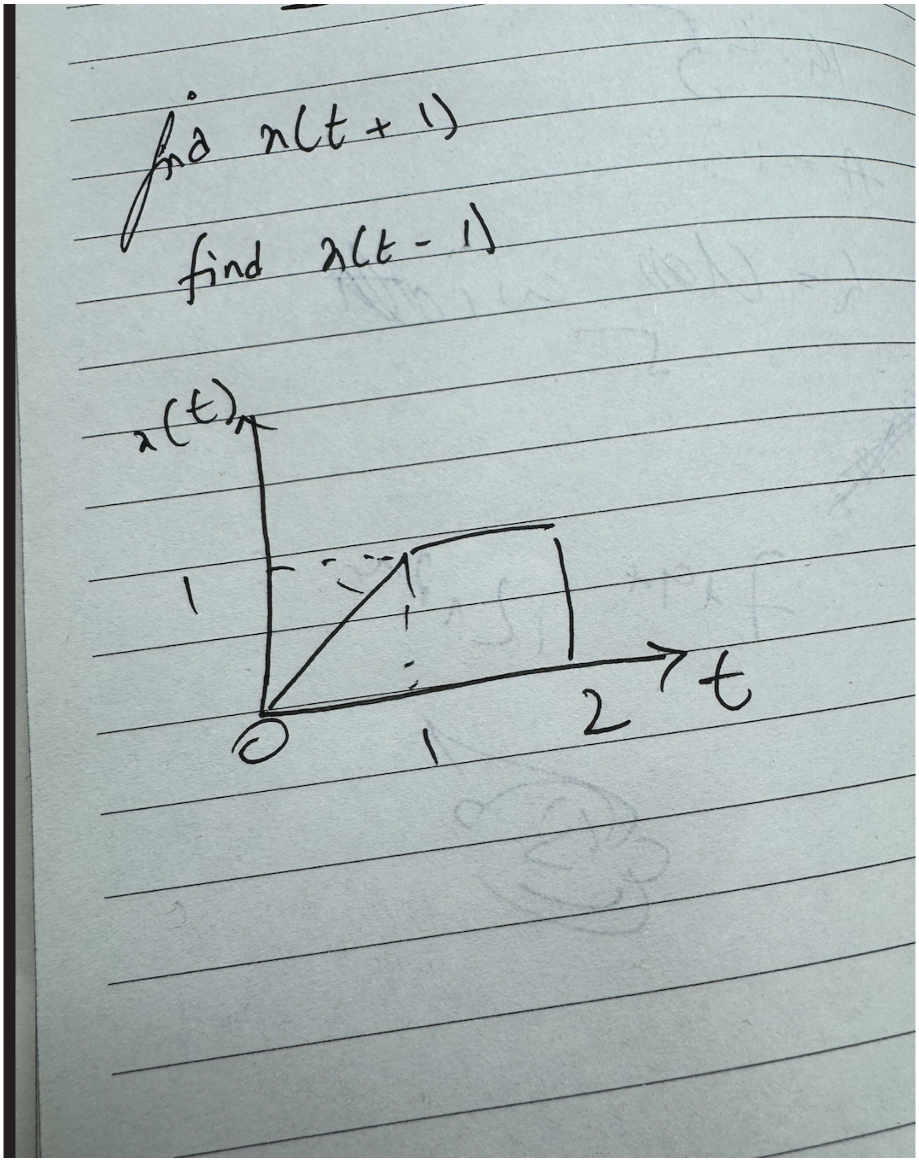
Trigger: Whenever addition or subtraction is involved inside the argument of $x(t)$, e.g. $x(t+1)$, $x(t-1)$.

Key property: During shifting, the **shape and amplitude of the signal remain unchanged** — only its position on the time axis changes.

How to perform a shift

1. Draw the graph and label the axes.
2. Equate the bracket term to `0` to find the shift point.

3. Solve for `t`.



Example — `x(t+1)`

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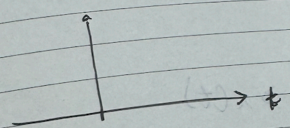
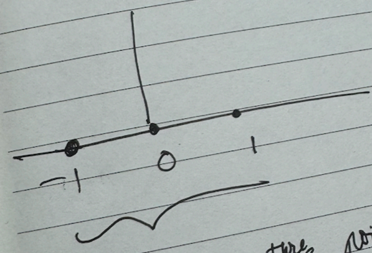
t + 1 = 0
t = −1
```

$x(t+1)$

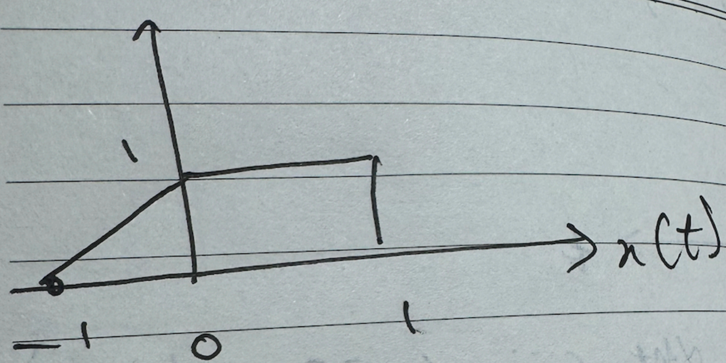


on Shifting to the right ($\because$ $t+1=0 \rightarrow t=-1$)



all the three points are shifted to the left.

$x(t-1)$



Ans

- The **negative sign** on `t` indicates a **left shift** (important).
- The magnitude (`1`) indicates the **unit** of shift.
- Shape and amplitude remain the same.

Time Delay vs Time Advance

Direction	Name	Sign convention
Shift **right**	**Time Delay**	Corresponds to `t - a` (positive shift)
Shift **left**	**Time Advance**	Corresponds to `t + a` (negative shift)

Intuition: A negative sign associated with the shift simply represents a **change in direction** (left vs right), not a change in shape/amplitude.

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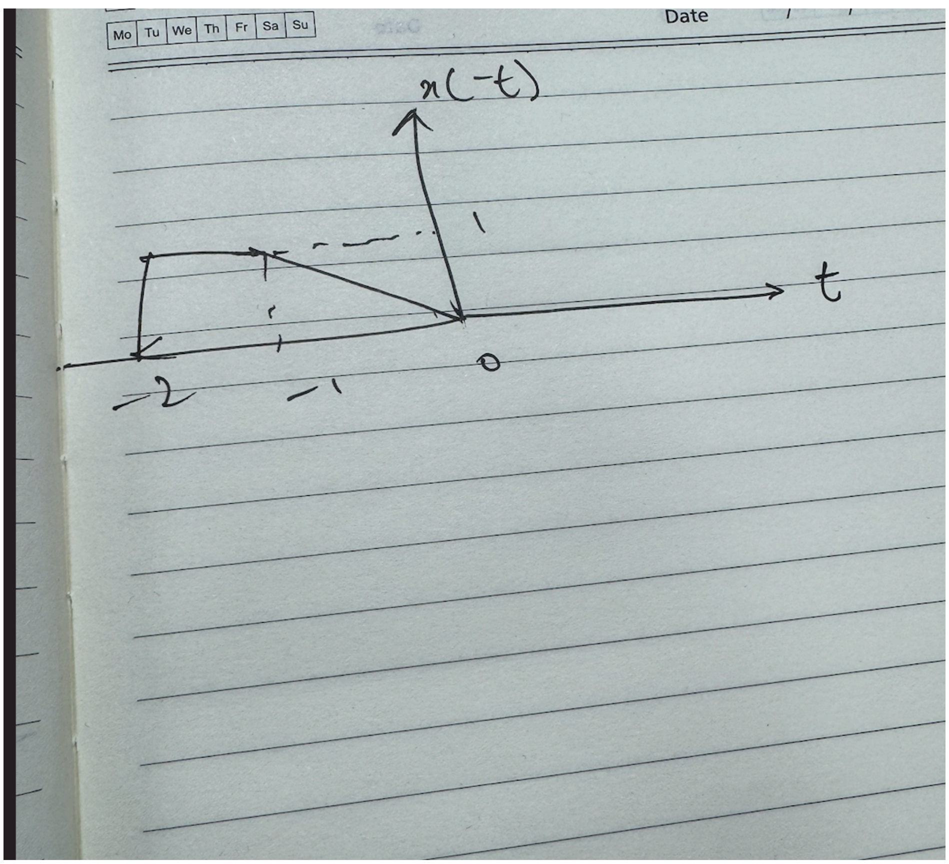
2. Folding Operation



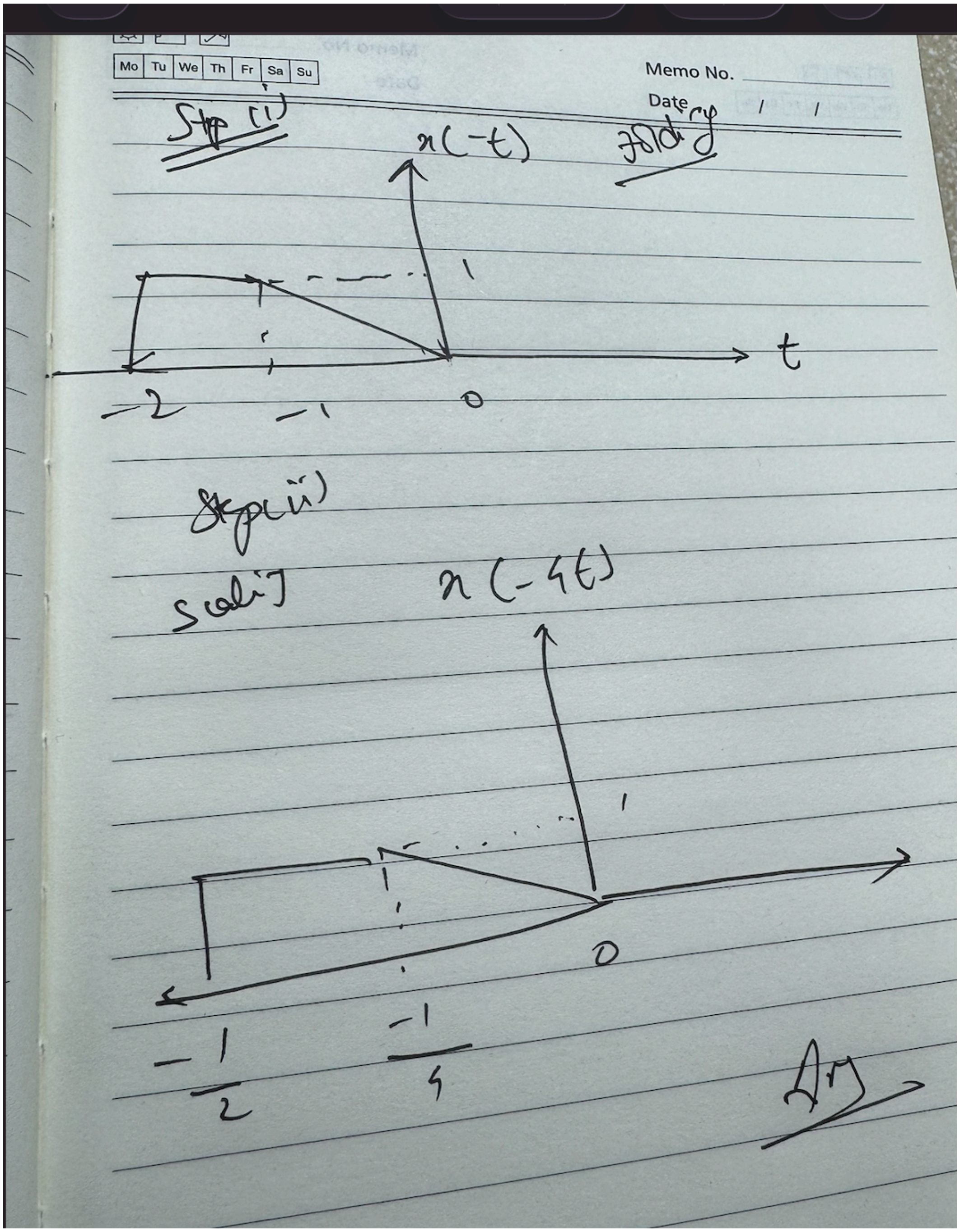
Trigger: Whenever `t` itself becomes **negative** inside the argument — i.e., `x(-t)`.

Key property: Folding produces a **mirror image of the signal about the y-axis** (symmetric reflection).

Rule of thumb

- `x(-kt)` → **Folding operation**
- `x(kt)` → **Scaling operation**

Example — `x(-4t)`



Since both a negative sign *and* a coefficient are present, apply operations in order:

1. **Fold first** → obtain `x(-t)`
2. **Then scale** → obtain `x(-4t)`

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3. Scaling Operation

Key property: Like shifting, scaling **preserves shape and amplitude** — it compresses/expands the signal along the time axis (or in the folding+scaling combo, also mirrors it).

Quick Reference Summary

Operation	Trigger in `x(·)`	Effect	Shape/Amplitude
Shifting	`t ± a` (addition/subtraction)	Moves signal left/right	Unchanged
Folding	`x(−t)` (negative `t`)	Mirrors signal about y-axis	Unchanged
Scaling	`x(kt)` (coefficient on `t`)	Compresses/expands signal	Unchanged

Combined operation order (e.g. `x(−kt)`): Fold → Scale